

AI Vision Analytics: System Documentation

Retail Intelligence Project

2026

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1 Introduction

The **AI Vision Analytics** system is a real-time retail intelligence platform designed to track customer behavior and emotional engagement. By leveraging Edge AI optimization through Intel’s **OpenVINO** toolkit, the system provides high-performance inference for person detection and emotion recognition directly in a web-based dashboard.

2 System Architecture

The system follows a modular architecture integrating computer vision, backend processing, and real-time frontend updates.

2.1 Core Components

- **Detection Engine:** Utilizes YOLOv8 for robust human detection and spatial tracking.
- **Emotion Recognition:** Powered by the OpenVINO optimized `emotions-recognition-retail` model.
- **Backend:** Flask-based Python server managing video streams and database transactions.
- **Real-time Communication:** WebSocket protocol (Flask-Sock) for sub-second telemetry updates.
- **Data Persistence:** SQLite3 database for historical visitor logging.

3 Technical Implementation

3.1 OpenVINO Optimization

To achieve high FPS (Frames Per Second), the emotion model is compiled for the CPU using the OpenVINO Inference Engine. The input undergoes a preprocessing stage:

$$Input_{tensor} = Transpose(Resize(Face_{ROI}, 64 \times 64)) \quad (1)$$

3.2 Database Schema

The system records every visitor session in a relational format:

4 Dashboard Interface

The user interface is designed as a single-page application (SPA) that fits entirely within the browser viewport ($100vh$).

Field	Type	Description
ID	INTEGER	Primary Key (Auto-increment)
Visitor_ID	INTEGER	Unique ID assigned by tracker
Entry_Time	TEXT	Timestamp of first detection
Exit_Time	TEXT	Timestamp after 30 unseen frames
Dominant_Emotion	TEXT	Statistical mode of detected emotions

Table 1: SQLite Visitor Table Schema

4.1 Visual Components

1. **Video Stream:** Processed frame-by-frame with bounding boxes. Using `object-fit: cover` ensures a full-bleed immersive view.
2. **Telemetry Cards:** Real-time count of total and active visitors.
3. **Emotion Distribution:** A Chart.js doughnut chart visualizing the aggregate sentiment.
4. **Live Log:** A scrollable table indicating active tracking IDs and their current emotional state.

5 Alerting System

The system includes an automated notification module. If a customer is detected with a sentiment of **Anger** or **Sadness**, the system triggers an SMTP alert:

- **Cooldown:** 60 seconds between alerts to prevent spam.
- **Security:** Credentials managed via `.env` environment variables.

6 Conclusion

This system demonstrates the efficiency of Edge AI in a retail environment. By combining modern web technologies with optimized AI models, it provides actionable insights into customer satisfaction without the need for high-latency cloud processing.